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Polyp-LVT: Polyp segmentation with lightweight vision transformers[☆,☆☆]

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ABSTRACT

Automatic segmentation of polyps in endoscopic images is crucial for early diagnosis and surgical planning of colorectal cancer. However, polyps closely resemble surrounding mucosal tissue in both texture and indistinct borders and vary in size, appearance, and location which possess great challenge to polyp segmentation. Although some recent attempts have been made to apply Vision Transformer (ViT) to polyp segmentation and achieved promising performance, their application in clinical scenarios is still limited by high computational complexity, large model size, redundant dependencies, and significant training costs. To address these limitations, we propose a novel ViT-based approach named Polyp-LVT, strategically replacing the attention layer in the encoder with a global max pooling layer, which significantly reduces the model's parameter count and computational cost while keeping the performance undegraded. Furthermore, we introduce a network block, named Inter-block Feature Fusion Module (IFFM), into the decoder, aiming to offer a streamlined yet highly efficient feature extraction. We conduct extensive experiments on three public polyp image benchmarks to evaluate our method. The experimental results show that compared with the baseline models, our Polyp-LVT network achieves a nearly 44% reduction in model parameters while gaining comparable segmentation performance.

1. Introduction

Colorectal cancer (CRC) ranks third among common cancers and is the second most deadly cancer worldwide [1]. Global statistics [2] for the year 2020 indicate that there were about 1.93 million new CRC cases and 0.94 million CRC caused deaths in that year worldwide. CRC typically develops from the progression of pre-colonic lesions/polyps through various oncogenic pathways [3]. Among these lesions, colonic polyps are the most common cause of CRC [4]. Regular screening, timely detection, and removal of pre-cancerous colorectal polyps at the early stage are crucial as they generally need at least over 10 years to evolve into malignancy [5], thereby preventing the development of CRC [6].

Colonoscopy is regarded as the gold standard for screening and early intervention in colorectal lesions. It could help doctors find and remove colorectal polyps before they turn into cancer [7]. However, as can be noticed from the images in Fig. 1, colon polyps always show large variations in shapes, sizes, and locations, and closely resemble the surrounding normal mucosal tissues in both texture and color. In addition, due to the artifacts caused by the continuous deformation of the colonic organs, food residues, and feces remaining after clearing the bowel, colorectal polyps are easily missed during colonoscopy, which leads to a high degree of dependence and subjectivity in colonoscopy procedures [8]. Therefore, developing a computer-aided clinical support system that can timely and accurately identify polyps in endoscopic images is a challenging task.

Traditional polyp segmentation mainly utilizes low-level features to extract polyps [9]. However, highly representative features need extensive prior domain knowledge of colorectal polyps. In addition, hand-crafted features manually designed by human experts have limited representation capability. Consequently, these methods based on hand-crafted features exhibit limitations in polyp segmentation due to the high intra-class variability and the low inter-class variations between polyps and hard mimics [10].

☆ Long Lin and Guangzu Lv contribute equally to the article.

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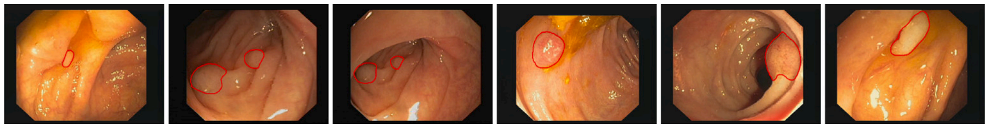

Fig. 1. Six example colon images. The region circled in red are polyps in each image.

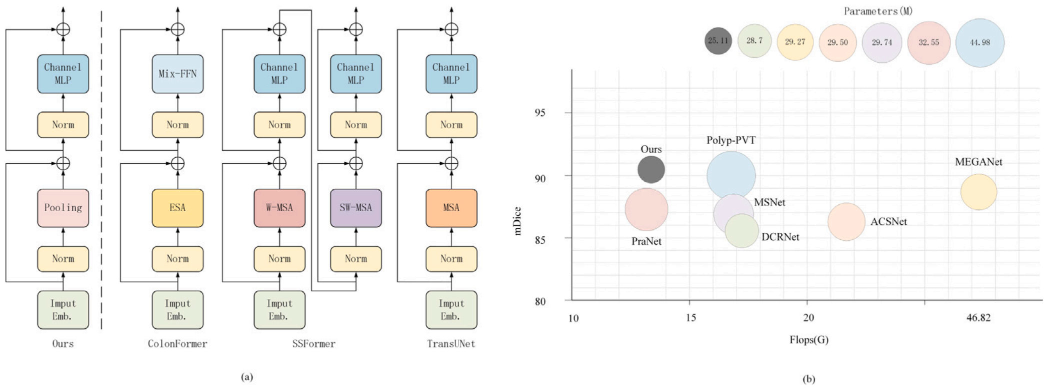

Fig. 2. The models presented in (a) underscore a fundamental divergence between our approach and the prevalent mainstream models. (b) Accuracy (mDice), Flops(G), and Parameters(M) comparison on Endoscene. Our approach achieves similar performance to Polyp-PVT [7] while reducing the parameters by almost 44% compared to Polyp-PVT [7].

Deep Convolutional Neural Networks(CNNs) have emerged as a powerful tool and revolutionized medical image segmentation in recent years [11–16]. Built upon the foundation of CNNs, the U-Net [11] architecture leverages a meticulously designed encoder–decoder structure to achieve remarkable effectiveness in medical image segmentation tasks. Variants of U-Net, including but not limited to UNet++ [12], RauNet [13], UNet 3+ [14], DcuNet [15], and SPU-Net [16], have also demonstrated highly competitive performance in this field. However, CNN-based models confront the limitations related to their receptive field, which makes it difficult to capture long-range dependencies between distant image features. This can be crucial for polyp segmentation where long-range dependencies are vital for accurate polyp identification. It also results in a partial loss of information, potentially impacting the segmentation accuracy of the results [17].

The Vision Transformer (ViT) [18] architecture effectively mitigates the inherent limitations of capturing long-range dependencies and has yielded favorable results within the polyp segmentation domain [7,19–23]. Polyp-PVT [7] employs a pyramid vision transformer backbone as an encoder to extract more robust and informative features from images. SSFormer [19] uses the hierarchical multiscale architecture of Swin Transformer to extract information from different layers. These ViT-based methods demonstrate their potential to becoming the leading technology in polyp segmentation. Nonetheless, ViT still grapples with challenges, including high computational complexity, large model size, redundant dependencies, and significant training costs. These challenges can hinder the rapid and precise segmentation of polyps in endoscopic images within computer-aided clinical support systems.

In order to obtain the rapid and precise segmentation of polyps within endoscopic images for computer-aided clinical support systems, this paper proposes a lightweight Vision Transformer named Polyp-LVT for polyp segmentation. Fig. 2(a) offers an overview of the Polyp-LVT's core network structure. Attention mechanisms in transformers are computationally expensive, and require a large number of parameters and significant computational resources. The proposed Polyp-LVT strategically replaces an attention layer in the encoder with a global max pooling layer, dramatically reducing model complexity and resource consumption of the model. Furthermore, we introduce an Inter-block Feature Fusion Module (IFFM) in the decoder, offering a streamlined yet highly efficient feature extraction mechanism. IFFM could extract both high-level and low-level image features and fuse them deeply to improve the precision of polyp identification. Fig. 2(b) shows that our model is efficient in parameter optimization. Impressively, our model achieves comparable segmentation performance compared to the popular SOTA methods while consuming merely 13.21 GFlops and 25.11 million parameters.

In summary, our contributions to this work can be summarized into four aspects:

- We design Polyp-LVT, a novel lightweight visual transformer, for rapid and accurate polyp segmentation in endoscopic images.
- Polyp-LVT introduces a strategic substitution in its encoder, swapping out an attention layer for a global max pooling layer. This modification significantly reduces the model's complexity while enhancing its computational efficiency.
- Polyp-LVT incorporates the Inter-block Feature Fusion Module (IFFM) into the decoder, providing a sleek and efficient mechanism for extracting features.
- Extensive experiments demonstrate that our approach achieves comparable performance while requiring only 13.21GFlops and 25.11M parameters (about a nearly 44% reduction to the baseline models).

The remainder of this paper is organized as follows: Section 2 provides a brief review of related work. In Section 3, we present the proposed Polyp-LVT model for polyp segmentation. Experimental results and analysis are given in Section 4. Finally, we conclude the paper in Section 5.

2. Related work

In this section, we briefly survey the methods used in the field of polyp segmentation. First, we give a summary of the traditional methods. Next, the CNN architecture and its variants, especially the U-Net model are given. Finally, we investigate the methods based on Vision Transformer.

2.1. Traditional approaches

Over the past two decades, researchers have tried to automatically segment and detect polyps in endoscopic images and provide an effective alternative to traditional manual detection [24]. The earlier methods [25,26] mainly focus on the use of hand-crafted features such as color, texture, border and so on for polyp segmentation.

Jerebko et al. [27] designed a method that combines the Canny edge detector and Radon transforms for polyp detection. This approach calculates the shape features including polyp base length, polyp boundary length, polyp height, polyp internal area, average intensity, and inscribed circle radius to detect the polyp region. Yoshida et al. [28] proposed a method for polyp detection based on some volumetric features computed at each location in the extracted colon. It adopts hysteresis thresholding and fuzzy clustering based on feature values to detect polyps.

Traditional approaches for polyp segmentation often rely on manually engineered features that concentrate on extracting low-level characteristics such as shape, color, and texture information to distinguish polyps from their surrounding tissues [9]. However, the substantial variations in color, morphology, and texture across different polyps render these hand-crafted features inadequate for comprehensive characterization [29]. Hampered by high rates of missed or erroneous detections, traditional methods have difficulty achieving clinically acceptable performance in complex scenarios with highly variable polyp attributes.

2.2. Convolutional neural networks based approaches

To overcome the limitations of hand-crafted feature descriptors, Convolutional Neural Networks (CNNs) based approaches tried to extract more powerful features to classify colonic polyp for colon cancer screening [30]. Brando et al. [31] adopted Fully Convolutional Networks(FCNs) [32] and a pre-trained VGG13 model [33] to extract effective deep features for the classification of polyp patches. Akbari et al. [34] proposed an end-to-end segmentation scheme based on the FCN-8S network. The proposed method generates a prediction map with the same size as the original image of the input network [35].

With the proposal of encoder–decoder based models, U-Net [11] and its variants [12–16,36–38] have been successfully applied to the field of polyp segmentation. U-Net performs jump concatenation of features extracted by the encoder with those extracted by up-sampling the decoder to produce the final segmentation prediction. U-Net++ [12] uses joint learning by sharing encoders so as to enhance the image features. However, because of the blurring of the boundary between the polyp and the intestinal wall, these methods do not deal well with polyp boundaries, which is essential for accurate polyp segmentation [39].

Unlike U-Net, PraNet [10] adopts a two-stage segmentation approach that utilizes parallel decoders to predict rough regions and an attentional mechanism to recover the edges and internal structure of the polyps for fine segmentation. Psi-Net [40] develops a structure consisting of one encoder and three parallel decoders, which facilitates the acquisition of shape and boundary information. All these methods greatly improve the perception of polyp boundaries with good results. In addition, ACS-Net [41] and HRE-Net [42] improve the prediction accuracy by paying more attention to the contextual information in order to reduce the ambiguity of local features. While these CNN based methods have shown success in polyp segmentation, their ability to capture long-range dependencies remains limited [43].

2.3. Vision transformer based approaches

Following the revolutionary impact of Transformers on NLP [44], the Vision Transformer (ViT) [18] has brought a similar revolution to many computer vision tasks. ViT divides the image into smaller patches, which are processed sequentially by a transformer encoder, and then uses the extracted information from the encoder to classify the image. This approach greatly reduces the computational cost and allows the transformer to handle large images. ViT is capable of capturing long-term dependencies between pixels and has achieved excellent results in the field of polyp segmentation. Polyp-PVT [7] uses pyramid-structured ViT to extract more powerful features that characterize spatial correlation and the global context of the image. Colonformer [17] adopts a novel network to address the lack of decoding process in the limitations of spatially dependent multilevel feature representations. PolypSeg [45] leverages adaptive scale and semantic global context modules to identify key regions and strengthen the integration of high-level and low-level information.

Although ViT has achieved excellent results, its training requires a large dataset [46]. This property hinders its application to problems such as medical image analysis, including polyp segmentation, where data is often scarce. For example, even the most extensive publicly available gastrointestinal dataset, Kvasir [47], has only 1,000 images and corresponding ground-truth annotations. Similarly, another public resource, the CVC-ClinicDB [48] dataset, has only 612 images. Because of the dataset limitation, this can easily lead to overfitting of the model. DeiT [49] introduces a data efficient training strategy combined with distillation methods, which is helpful for improving the performance when training on small datasets. BoxPolyp [50] utilizes accurate masking and additional coarse frame annotations to mitigate the overfitting problem observed in previous polyp segmentation models. SSFormer [19] adopts a simple multilayer perceptron decoder to adjust the pyramid transformer backbone, emphasizing local features and limiting distraction, thus addressing the overfitting challenge.

In addition, ViT suffers from a large number of parameters, a long training time, and a slow training speed. TransUNet [51] uses a transformer-based network with a hybrid ViT encoder and an up-sampled CNN decoder. The hybrid ViT stacks CNNs and transformers together, resulting in higher computational cost. TransFuse [52] solves this problem by using a parallel architecture. Both TransUNet [51] and TransFuse [52] use an attention gate mechanism and a so-called dual fusion module. These components make the network architecture large and highly complex.

To overcome this limitation, researchers have also proposed various approaches. Segformer [53] decreases the computational complexity by improving the self-attention mechanism. PVT [54] reduces the cost by using a spatially reduced attention mechanism. PVTv2 [55] improves the performance by incorporating the current complex attention layer and a convolutional feedforward network.

As research progresses, recent developments have demonstrated that the attention-based module in Vision Transformers can be effectively substituted with spatial multilayer perceptrons (MLPs). In the context of Metaformer [56], a straightforward spatial pooling operator has been introduced as a replacement for the attention-based module, and it has surprisingly exhibited remarkable performance. This paper delves into the potential of pooling operators for designing an efficient network. Within this study, we adhere to the structural framework of the transformer model and introduce an effective pooling layer to replace the conventional attention mechanism. Additionally, we propose the IFFM to facilitate image feature extraction through multiple layers. Our model demonstrates exceptional performance with resource efficiency, requiring only 13.21 GFlops and 25.11M parameters to achieve outstanding results.

3. Method

3.1. Overall architecture

As shown in Fig. 3, our proposed Polyp-LVT mainly consists of two modules: a transformer encoder based on the pyramid structure and a specially designed module for polyp segmentation named Inter-block Feature Fusion Module (IFFM).

3.2. Transformer encoder

Due to the interference of the environment and equipment during the acquisition process, there is a lot of noise in the image. In addition, polyps have significant differences in the size, shape, and boundary of the polyp image, making the segmentation task more challenging.

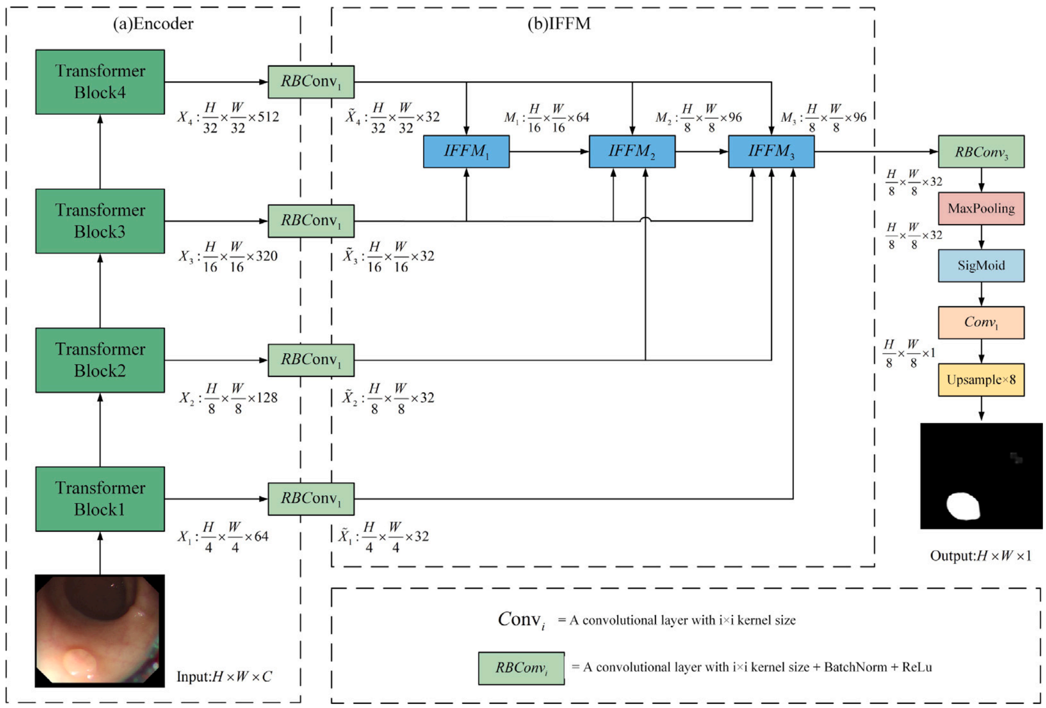

Fig. 3. The architecture of our proposed Polyp-LVT. It comprises two modules: (a) Transformer encoder, (b) Inter-block Feature Fusion Module (IFFM). The transformer encoder uses a PVTv2 encoder to extract features. The details of the transformer block are shown in Fig. 4. IFFM follows a three-stage workflow as illustrated in Fig. 5.

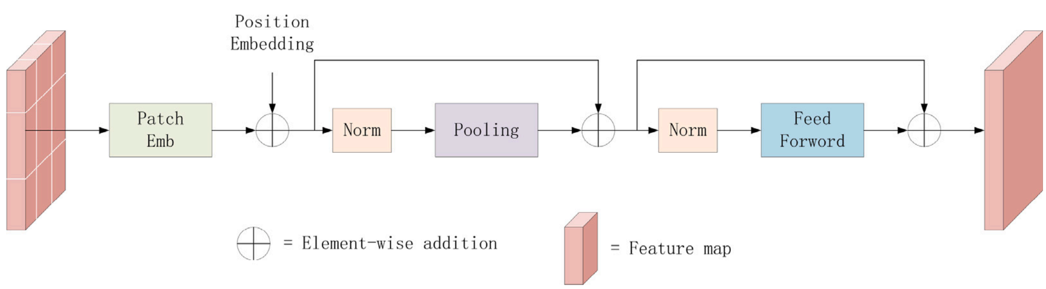

Fig. 4. Illustration of a transformer encoder block in Fig. 3(a). Inspired by the MetaFormer [56], we use a global max pooling layer instead of the attention operation in the transformer.

Recent research [54] has shown that the Transformer based on the pyramid structure exhibits strong performance and robustness in polyp segmentation. Therefore, this paper adopts the PVTv2 [54] based on the pyramid structure as the backbone network of the encoder. The advantage of this structure is that it can reduce resource consumption while better extracting image features [57]. However, the attention block in PVTv2 is a weight-intensive module, increasing the number of parameters. Therefore, we replace the attention block in PVTv2 with a global max pooling block [56], which further reduces the model parameters and significantly lowers resource consumption.

As shown in Fig. 3(a), the transformer encoder is divided into four stages. The number of blocks from the first stage to the fourth stage is 3, 4, 18, and 3, respectively. The input original image is $X \in \mathbb{R}^{H\times W\times 3}$, and the resolution of the layered characteristic figure $X_i \in \mathbb{R}^{\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}\times Channel(i)}$, where $i \in \{1,2,3,4\}$, and $Channel(i)$ is defined as follows:

$$Channel(i) = \begin{cases} 2^{5+i}, & i \in \{1,2,4\} \\ 320, & i = 3. \end{cases} \tag{1}$$

3.3. Inter-block Feature Fusion Module (IFFM)

In order to acquire global pixel-level polyp features, we employ a multi-level feature fusion approach. The entire module is composed of three cascaded components. It should be noted that, due to the characteristics of the pyramid structure encoder, the channel numbers of the feature maps generated at different layers are diverse. To facilitate fine-grained fusion, we use a convolutional block named $RBConv_p$ to keep the channel of the obtained feature maps X_i into 32. The entire computational process can be formulated as follows:

$$RBConv_p(\cdot) = ReLu(BN(Conv_p(\cdot))), \tag{2}$$

where $Conv_p$ is a convolutional layer with $p\times p$ kernel size, $ReLu(\cdot)$ is the ReLU activation layer, $BN(\cdot)$ represents batch normalization. Under the condition $p = 1$, we employ the $RBConv_1$ operator on the input vectors X_1 through X_4, resulting in transformed outputs denoted as $\tilde{X}_1$ through $\tilde{X}_4$. This transformation can be formally expressed as:

$$\tilde{X}_i = RBConv_1(X_i), \tag{3}$$

where $i \in \{1,2,3,4\}$.

All $\tilde{X}_i$ are fed into IFFM. $RBConv_3$ adopts a 3×3 convolution kernel and is widely used in IFFM. For simplicity, we use $\mathcal{R}(\cdot)$ to represent $RBConv_3(\cdot)$. As shown in Fig. 3, the IFFM mainly consists of three stages, as follows:

(1) In the first stage, as shown in Fig. 5(a), the highest-level feature map $\tilde{X}_4$ is upsampled by a factor of two to match the resolution of the third-level feature map $\tilde{X}_3$. Subsequently, it is fed through $\mathcal{R}(\cdot)$,

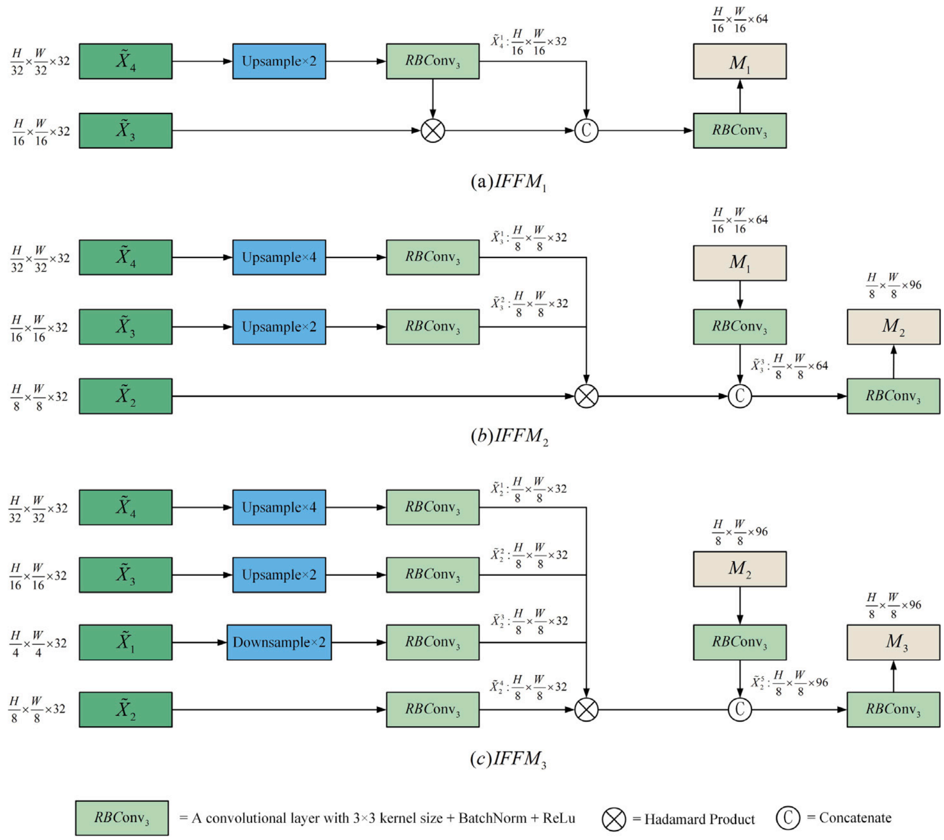

Fig. 5. The details of three stages of IFFM module: (a) $IFFM_1$, (b) $IFFM_2$, (c) $IFFM_3$.

outputting $\tilde{X}_4^1$. Afterward, the product of $\tilde{X}_4^1$ and the feature map $\tilde{X}_3$ is computed, and the obtained result is concatenated with $\tilde{X}_4^1$, leading to the feature map. Finally, we use $\mathcal{R}(\cdot)$ to fuse the concatenated features, yielding fused feature map $M_1 \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{16}\times 64}$. The entire computational process can be formulated as follows:

$$M_1 = \mathcal{R}(C(\mathcal{R}(U_2(\tilde{X}_4)) \odot \tilde{X}_3, \mathcal{R}(U_2(\tilde{X}_4)))), \tag{4}$$

where $C(\cdot)$ denotes the concatenation operation, $\odot$ signifies the Hadamard product, $U_2(\cdot)$ represents the double up-sampled.

(2) In the second stage, as shown in Fig. 5(b), the feature map $\tilde{X}_4$ is from the fourth layer, and the feature map $\tilde{X}_3$ is from the third layer. They are upsampled by a factor of four and two respectively, to match the size of the feature map $\tilde{X}_2$ from the second layer. Subsequently, the upsampled feature maps are respectively input into $\mathcal{R}(\cdot)$, producing $\tilde{X}_3^1$ and $\tilde{X}_3^2$. Following this, the products of $\tilde{X}_3^1$, $\tilde{X}_3^2$, and the second-layer feature map $\tilde{X}_2$ are computed, and the resulting outputs are concatenated with $\tilde{X}_3^3$. Finally, we use $\mathcal{R}(\cdot)$ to fuse the concatenated features, yielding fused feature map $M_2 \in \mathbb{R}^{\frac{H}{8}\times\frac{W}{8}\times 96}$. The entire computational process is formulated as Eq. (5):

$$M_2 = \mathcal{R}(C(\mathcal{R}(U_4(\tilde{X}_4)) \odot \mathcal{R}(U_2(\tilde{X}_3)) \odot \tilde{X}_2, \mathcal{R}(U_2(M_1)))), \tag{5}$$

where $U_4(\cdot)$ represents the quadruple up-sampled.

(3) In the third stage, as shown in Fig. 5(c), the feature maps $\tilde{X}_4$, $\tilde{X}_3$, and $\tilde{X}_1$ from the fourth, third, and first layers are respectively upsampled by factors of 4, 2, and 0.5, to achieve the same resolution as the second-layer feature map $\tilde{X}_2$. Subsequently, the upsampled $\tilde{X}_4$, $\tilde{X}_3$, $\tilde{X}_2$ along with the feature map M_2 from the previous stage are passed through $\mathcal{R}(\cdot)$, generating results $\tilde{X}_2^1$, $\tilde{X}_2^2$, $\tilde{X}_2^3$, $\tilde{X}_2^4$ and $\tilde{X}_2^5$, respectively. Following this, the products of $\tilde{X}_2^1$, $\tilde{X}_2^2$, $\tilde{X}_2^3$ and $\tilde{X}_2^4$ are computed, and the resulting outputs are concatenated with $\tilde{X}_2^5$. Finally, we use $\mathcal{R}(\cdot)$ to fuse the concatenated features, yielding fused feature map $M_3 \in \mathbb{R}^{\frac{H}{8}\times\frac{W}{8}\times 96}$. The entire computational process is formulated as Eq. (6):

$$M_3 = \mathcal{R}(C(\mathcal{R}(U_4(\tilde{X}_4)) \odot \mathcal{R}(U_2(\tilde{X}_3)) \odot \mathcal{R}(\tilde{X}_2) \odot \mathcal{R}(D_2(\tilde{X}_1)), \mathcal{R}(M_2))), \tag{6}$$

where $D_2(\cdot)$ represents the double down-sampled.

After obtaining the feature map $M_3 \in \mathbb{R}^{\frac{H}{8}\times\frac{W}{8}\times 96}$, we then perform the following operations to obtain the predicted image.

$$OutPut = U_8(Conv_1(S(MaxPooling(\mathcal{R}(M_3))))), \tag{7}$$

where $U_8(\cdot)$ represents the eight times up-sampled operator, $Conv_1(\cdot)$ denotes the 1×1 convolution, and $S(\cdot)$ denotes the sigmoid function.

3.4. Loss function

Polyp-LVT uses a combination of weighted intersection over union (mIoU) loss [58], log-cosh dice (LCD) loss [59], and weighted binary cross entropy (BCE) loss [58,59] to train the model.

The loss function is designed to address both global and local structural aspects. It often involves two loss functions: a global one that focuses on the object level and a local one that operates at the pixel level. Specifically, we use mIoU loss and LCD loss to consider the global structure. For the local structure, we adopt BCE loss, which handles each pixel separately during optimization.

Since different regions of pixels contribute differently to the model-learning process. We denote the importance of the pixel (i, j) by the weight β_{ij}. As suggested in [58], the weight β_{ij} of a pixel (i, j) is defined as the difference between the center pixel and its neighbors:

$$\beta_{ij} = \left| \frac{\sum_{m,n \in \mathbb{N}_{ij}} g_{mn}}{|\mathbb{N}_{ij}|} - g_{ij} \right|, \tag{8}$$

where $\mathbb{N}_{ij}$ denotes a 31×31 neighborhood of pixel (i,j), and $g_{ij} \in \{0,1\}$ is the true label of pixel (i,j).

The weighted mIoU loss is defined as follows:

$$\mathcal{L}^{w}_{\text{mIoU}} = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}(g_{ij} \times p_{ij}) \times (1+\lambda\beta_{ij})}{\sum_{i=1}^{H}\sum_{j=1}^{W}(g_{ij} + p_{ij} - g_{ij} \times p_{ij}) \times (1+\lambda\beta_{ij})}, \quad (9)$$

where p_{ij} represents the predicted probability of pixel (i,j) belonging to the polyp category, and λ is a hyperparameter used to adjust the influence of the importance weight β_{ij}.

The log-cosh dice loss is defined as follows:

$$\mathcal{L}_{\text{LCD}} = log(cosh(1 - \frac{2 \times \left|p_{ij} \cap g_{ij}\right|}{|p_{ij}| + |g_{ij}|})), \quad (10)$$

where the cosh function is a hyperbolic cosine function and is defined as:

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2}. \quad (11)$$

The cosh function calculates the predicted distance from boundary pixels to the true boundary. The smaller the distance, the closer the cosh function output is to 1, indicating that the pixel is closer to the true boundary.

Following a common practice in training models, we also utilize the widely-used weighted binary cross entropy as the loss function. This function is defined mathematically as follows:

$$\mathcal{L}^{w}_{\text{BCE}} = -(\beta_{ij} \times p_{ij} \times log(g_{ij}) + (1-p_{ij}) \times log(1-g_{ij})). \quad (12)$$

The total loss of our model Polyp-LVT is calculated as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}^{w}_{\text{mIoU}} + \mathcal{L}_{\text{LCD}} + \mathcal{L}^{w}_{\text{BCE}}. \quad (13)$$

4. Experiments and discussions

4.1. Datasets and baselines

CVC-ClinicDB dataset [48]: This dataset is built from 31 colono scopy videos, containing a total of 612 images of size 384 × 288.

Kvasir-SEG dataset [47]: The dataset includes 1000 images with different resolutions between 720 × 576 and 1920 × 1072 pixels.

Endoscene dataset [60]: This dataset contains 60 images derived from endoscopic video analysis.

Setting: To ensure the fairness of the experiment, we follow the same protocol as used in PraNet [10] for our training process. The training set comprises 900 images from the Kvasir-SEG dataset and 548 images from the CVC-ClinicDB dataset. The remaining images from the Kvasir-SEG and CVC-ClinicDB datasets, as well as the Endoscene dataset, are respectively used to construct three test sets.

Baselines: For performance comparison, the following state-of-the-art polyp segmentation models are used as baselines: U-Net [11], DCRNet [61], ACSNet [41], MSNet [62], PraNet [10], Polyp-PVT [7], and MEGANet [63].

4.2. Evaluation metrics

Several standard evaluation metrics, including Dice [64], IoU, E-measure (E_{ξ}) [65,66], and mean absolute error (MAE), are used to assess the performance of the model. For the first three metrics, larger values correspond to better performance. In contrast, the last metric prioritizes lower values.

In the context of image segmentation tasks, the Dice and IoU metrics are used to measure how well predicted results match the ground truth. In the experiments, we use the average values of Dice and IoU, which are denoted as mDice and mIoU, respectively. The E-measure (E_{ξ}) quantifies the image segmentation by calculating a score based on how closely the predicted object outlines to match the ones in the ground truth data. We evaluate the model using the mean of (E_{ξ}), denoted as mE_{ξ}, which is adopted in the experiment.

Table 1
The hyperparameter setting for our Polyp-LVT model.

Optimizer	AdamW	Learning rate (lr)	3e−5
Multi-scale	[0.75, 1, 1.25]	Clip	0.5
Decay rate	0.1	Batch size	16
Weight decay	3e−5	Depth	[3, 4, 18, 3]
Epochs	150	Input size	352 × 352

Table 2
The comparative results on the CVC-ClinicDB dataset.

Model	mDice	mIoU	mE_{ξ}	MAE
U-Net [11]	0.823	0.755	0.913	0.019
DCRNet [61]	0.896	0.844	0.964	0.010
ACSNet [41]	0.882	0.826	0.947	0.011
MSNet [62]	0.921	0.879	0.972	0.008
PraNet [10]	0.899	0.849	0.963	0.009
Polyp-PVT [7]	0.937	0.889	0.985	0.006
MEGANet [63]	0.930	0.885	0.980	0.008
Ours	0.935	0.882	0.982	0.007

Table 3
The comparative results on the Kvasir-SEG dataset.

Model	mDice	mIoU	mE_{ξ}	MAE
U-Net [11]	0.818	0.746	0.881	0.055
DCRNet [61]	0.886	0.825	0.933	0.035
ACSNet [41]	0.898	0.838	0.941	0.032
MSNet [62]	0.907	0.862	0.944	0.028
PraNet [10]	0.898	0.840	0.944	0.030
Polyp-PVT [7]	0.917	0.864	0.956	0.023
MEGANet [63]	0.911	0.859	0.954	0.026
Ours	0.909	0.851	0.941	0.024

Table 4
The comparative results on the Endoscene dataset.

Model	mDice	mIoU	mE_{ξ}	MAE
U-Net [11]	0.710	0.627	0.847	0.220
DCRNet [61]	0.856	0.788	0.943	0.010
ACSNet [41]	0.863	0.787	0.939	0.013
MSNet [62]	0.869	0.787	0.943	0.010
PraNet [10]	0.871	0.797	0.950	0.010
Polyp-PVT [7]	0.900	0.833	0.973	0.007
MEGANet [63]	0.887	0.818	0.959	0.009
Ours	0.904	0.835	0.967	0.006

4.3. Implementation details

There are two hyperparameters including optimal learning rate and model depth, which should be predetermined. Their different choice would affect the performance of the model. In Figs. 6 and 7, we separately show the model performance under their different settings. As can be seen that setting the learning rate to 3e−5 and the model depth in [3, 4, 18, 3] is better than other settings. Our experiment is conducted using the PyTorch framework. We employ the AdamW optimizer to update the model parameters and train the model for 150 epochs with a batch size of 16. The training process is executed on an NVIDIA RTX 3090 GPU with 24 GB of memory. The complete network training duration amounts to approximately 8 h. The detailed experimental settings are summarized in Table 1.

4.4. Performance comparisons

We evaluate the performance of the proposed model by comparing it with the benchmark models. Table 2 and Table 3 respectively compare the performance of our Polyp-LVT model with the benchmark methods

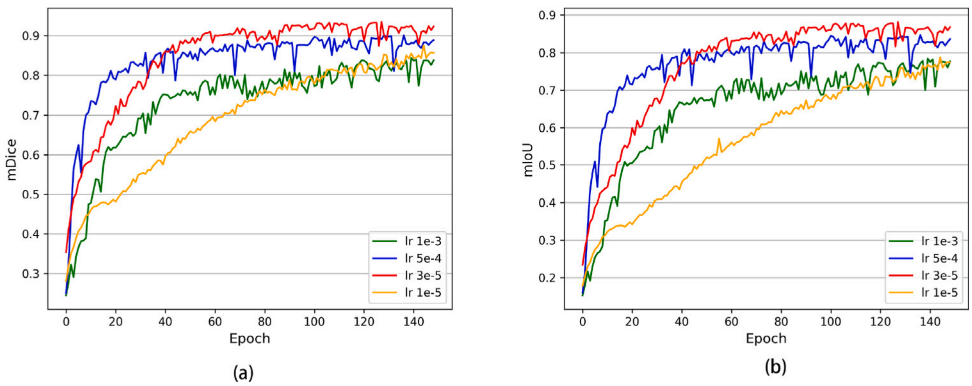

Fig. 6. The performance results of our model under different learning rates(lr): (a) for mDice. (b) for mIoU.

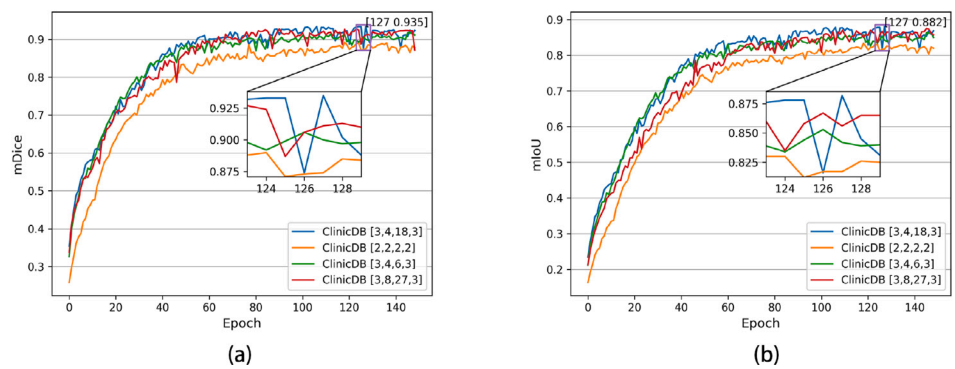

Fig. 7. The scores of two metrics, (a) mDice, and (b) mIoU, achieved by our model at different depths. The local zoom-in shows the best one of the several cases.

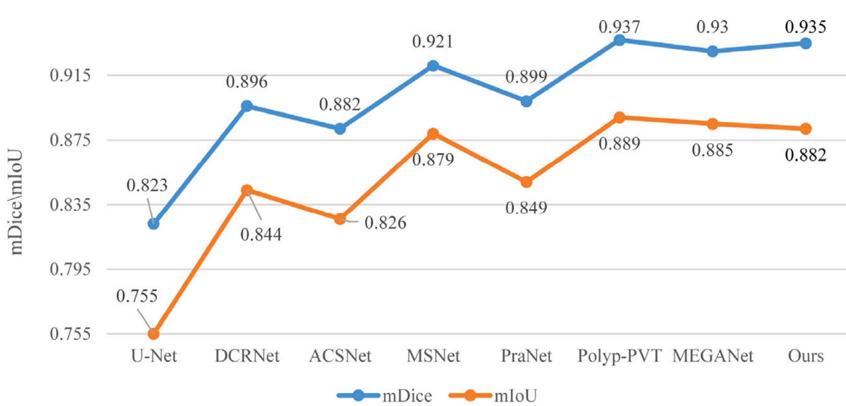

Fig. 8. Comparison of our model with other benchmark models on the CVC-ClinicDB dataset.

on the CVC-ClinicDB and Kvasir-SEG datasets. The experimental results indicate that the proposed Polyp-LVT demonstrates excellent performance across numerous metrics. Specifically, on the CVC-ClinicDB dataset, our model achieves comparable results to the state-of-the-art Polyp-PVT model and outperforms other approaches. On the mDice metric, Polyp-LVT exceeds MEGANet, MSNet, and PraNet by 0.5%, 1.4%, and 3.6%, respectively. On the Kvasir-SEG dataset, our model is slightly lower than Polyp-PVT but still reaches an imposing level, surpassing PraNet, MSNet, and ACSNet by 1.1%, 0.2%, and 1.1%, respectively, on the mDice measure.

Table 4 demonstrates the performance of Polyp-LVT on the Endoscene dataset. Compared to other well-performing methods, the proposed Polyp-LVT gains the best results across all four metrics. Specifically, on the mDice metric, Polyp-LVT exceeds the top-performing Polyp-PVT, MEGANet, and PraNet models by 0.4%, 1.7%, and 3.3%, respectively. On the mIoU metric, our model surpasses these three methods by 0.2%, 1.7%, and 3.8%, respectively. Fig. 8 provides a visual comparison of our model with other benchmark models on the CVC-ClinicDB dataset. It can be seen that our model achieves comparable performance with other benchmark models.

The qualitative results of the experiments are shown in Fig. 9. GT represents the ground truth segmentation of the images. In the first and last images, our model clearly performs better than other models in handling situations where the polyp and its surrounding environment is relatively similar, and the boundaries are blurred. Particularly in the last image, our segmentation result is very close to the GT, while the other models exhibit larger discrepancies. The qualitative results in Fig. 9 reveal that the segmentation performance of our Polyp-LVT model is superior to the other models.

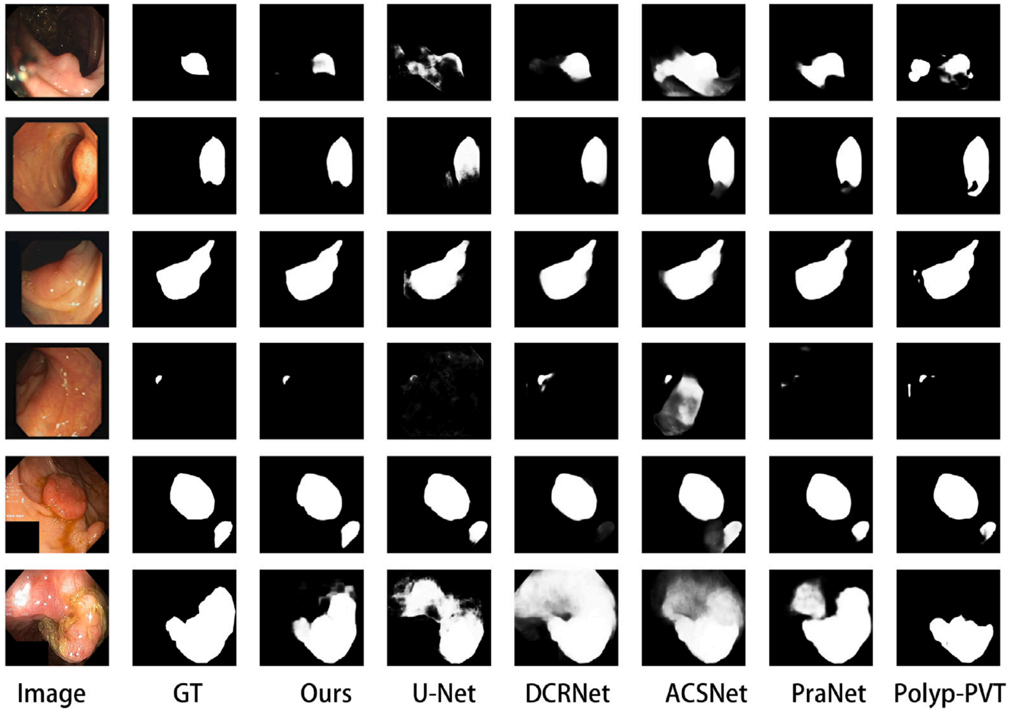

Fig. 9. Visualization results with the benchmark models which show that the proposed model is able to accurately locate and segment polyps regardless of their location, shape, and size.

Table 5
The comparisons of various models on the parameter sizes and Flops.

Model	Type	Parameters(M)	Flops (G)
DCRNet [61]	CNN	28.70	17.25
ACSNet [41]	CNN	29.50	21.61
MSNet [62]	CNN	29.74	16.97
PraNet [10]	CNN	32.55	13.11
Polyp-PVT [7]	Transformer	44.98	16.95
MEGANet [63]	CNN	29.27	46.82
Ours	Poolformer	25.11	13.21

Table 6
Ablation study on the effectiveness of different components in the IFFM module.

Components			CVC-ClinicDB		Kvasir-SEG		Endoscene	
$IFFM_1$	$IFFM_2$	$IFFM_3$	mDice	mIoU	mDice	mIoU	mDice	mIoU
–	✓	✓	0.929	0.873	0.903	0.845	0.895	0.826
✓	–	✓	0.915	0.853	0.901	0.842	0.887	0.810
✓	✓	–	0.931	0.879	0.905	0.847	0.894	0.822
✓	✓	✓	**0.935**	**0.882**	**0.909**	**0.851**	**0.904**	**0.835**

4.5. Comparisons of model efficiency

To verify the model's efficiency in terms of parameter count and computational resources, two metrics, model parameters and Floating-Point Operations per Second (Flops), are used in our experiments. Flops are an indicator of the model's computational capability, and a lower Flops value suggests that the model requires fewer computational resources.

Table 5 summarizes the comparison of the parameters and the Flops among various benchmark methods. The results show that the number of parameters in our model is reduced by almost 44% compared to Polyp-PVT. Compared with the lightest CNN model or the model based on transformer structure, our model has the lowest number of parameters and is at a low level in Flops. This also implies that our model requires fewer resources and operates in higher efficiency during inference and deployment.

4.6. Ablation experiments

In this subsection, we examine the effectiveness of different components of the IFFM module in our model. The same experimental environment and settings are taken in our experiments. The results are summarized in Table 6.

Effectiveness of $IFFM_1$: To analyze the effectiveness of the topmost module, we directly fuse the feature maps from the two lower layers. Compared to the standard three-layer cascade, this approach shows a decrease in performance on the three datasets. Specifically, on the CVC-ClinicDB dataset, the mDice and mIoU metrics decreased by 0.6% and 0.9%, respectively.

Effectiveness of $IFFM_2$: We discuss the impact of intermediate layers on the accuracy of the model. We fuse the feature maps from the upper and lower layers. On all three datasets, the performance is lower compared to the normal cascade approach. In particular, on the Endoscene dataset, the mDice and mIoU metrics decreased by 1.7% and 2.5%, respectively.

Effectiveness of $IFFM_3$: We evaluate the influence of the bottom-most cascade on the model's accuracy. By combining the feature maps from the two upper layers, we find that the model's performance on all metrics is inferior to that of the standard model on all three datasets. Particularly, on the Endoscene dataset, the mDice score decreases from 0.904 to 0.894.

5. Conclusion

In this paper, we present a new deep model, named Polyp-LVT, for automatic polyp segmentation. It is based on the Vision Transformer (ViT), which has shown promising performance in various medical image segmentation tasks. For alleviating the issue of the cumbersome and high computational complexity of ViT and pursuing a lightweight model to meet the requirements of practical polyp segmentation applications, we introduce a global max pooling layer in place of the attention layer in the encoder to reduce the scale of the model parameters without degrading the model performance of ViT. We also introduce an inter-block feature fusion module (IFFM) into the decoder for streamlined and efficient feature extraction which facilitates the rapid and accurate identification of polyps. We have conducted extensive experiments on the public polyp segmentation benchmarks. The experimental results show that our proposed Polyp-LVT equipped with only 13.21 GFlops and 25.11M model parameters (about a nearly 44% reduction to the baseline models) achieves a comparable segmentation performance with the baseline models.

It is worth pointing out that although our proposed model has achieved promising performance improvement compared to the baselines, particularly in parameter count and computational resource consumption, its segmentation accuracy may have a certain distance from the high-precision segmentation requirements in some real clinical scenarios. So, how to further improve the segmentation accuracy of our Poly-LVT model as much as possible without increasing the parameters and computational overhead will be the focus of our future research on polyp segmentation.

CRediT authorship contribution statement

Long Lin: Writing – original draft, Validation, Software. **Guangzu Lv:** Writing – original draft, Validation, Software, Conceptualization. **Bin Wang:** Writing – review & editing, Formal analysis. **Cunlu Xu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Jun Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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